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RESEARCH

Combined use of BWM-TOPSIS methods in the selection of thermal power plant installation site in the Karapinar/Turkiye Region, at risk of sinkhole formation

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Abstract With the rapidly increasing world population and the need for industrialization, energy supply has become an important global problem. A significant part of the world's energy needs is provided by fossil fuels. About half of all global coal deposits are low-quality coals, including lignite. Karapınar/Konya, also the study area, has Turkiye's second richest lignite reserve. The region's lignite reserve can be used in thermal power plants for electricity generation in terms of its nature and the amount is an important opportunity to meet the energy demand for both region and country. The region contains many sinkholes, and the potential for the formation of new sinkholes makes the site selection for thermal power plants in the region an even more strategic decision. This study aims to propose the most suitable thermal power plant site for the region by using Multi-Criteria Decision Making methods and Geographic Information Systems in an integrated way. Within the scope of the study, a total of twelve sub-criteria were taken into consideration under the main criteria of Geological, Economic and Environmental. The Best–Worst Method was applied to determine the criteria weights, and by using the weights, a suitability map for the thermal power plant installation site was produced and candidate regions were determined. TOPSIS was applied to determine the most suitable location among the candidate regions. The Candidate Region in the easternmost part of Karapinar district was chosen as the most suitable site for the thermal power plant installation.

Keywords Sinkhole formation · Site selection · Thermal power plant · BWM · Multi-Criteria Decision Making · TOPSIS · GIS

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Introduction

Problems in the global energy sector are at their greatest level in almost 50 years after the energy crisis of the 1970s (BP, 2022). With its important role in economic growth, energy is an indispensable element of development programs and is critical for the sustainability of the modern economy (Bahar, 2005; Bayramoglu, 2022; TUBITAK, 1998). In recent years, when the future of energy is uncertain, the rapid increase in the world population has led to a rapid increase in energy consumption. For example, world primary energy consumption increased approximately 1.73 times per year in the 50-year period after

1970 (Coal Sector Report Lignite, 2022; BP, 2022). The COVID-19 pandemic has created a major crisis in countries around the world, affecting people's lives and livelihoods, and this crisis has caused the world's energy future to remain uncertain (World Energy Outlook, 2020). Another recent example of the problems in the sector is the events in Ukraine (BP, 2022). Technological developments, industrialization and the increase in the world population together with the latest events increase the energy demand, which is the main input in production, and cause the welfare levels of societies to increase (Koc & Kaya, 2015). Therefore, having energy resources, producing energy from these resources and keeping the transportation routes that will deliver the energy to the market under control are among the issues that states give importance to in foreign policy (Aksoy, 2016).

Today, with the increase in the population of countries around the world, a country's ability to meet its energy needs is now considered equivalent to independence (Yilankirkan & Dogan, 2020). Especially for a geopolitically important country like Turkiye, meeting its own energy needs has a key role in reducing foreign dependency and economic development. Electricity consumption in Turkiye continues to increase every year, reaching 327 TWh in 2021 with a growth of 9% and the industry ranks first in electricity consumption with 44%. Electricity generation was 329 TWh in 2021, with natural gas accounting for 33% and imported coal and lignite power plants accounting for 30% (Energy Sectoral Overview, 2022). While the total number of power plants in Turkiye was 11,427 as of December 2022, the total installed power was 103,809,3 MW. There are 46 lignite-fired power plants and the power produced is 10.191.5 MW. The power produced per power plant is 221.55 MW on average, which ranks third after imported coal and asphaltite (URL-1). The fact that the total power produced is high compared to the number of power plants indicates the potential for energy utilization. As a country with a rapidly increasing population, the General Directorate of Mineral Research and Exploration (MTA) has accelerated its efforts in recent years to meet the energy demand (MTA, 2012). As a result of exploration activities for lignite coal, which has an important role in mining, the apparent lignite reserve in Turkiye, which was reported 14.1 billion tons in 2013, was increased to 18.9 billion tons in 2018 (MTA, 2012). Lignites are mostly used as fuel in thermal power plants since the heating values of the lignites detected in Turkiye are quite low and more than half of the lignites in the country contain more than 20% moisture (EÜAŞ, 2016).

Konya-Karapinar region is the second richest region of Turkiye in terms of lignite reserves and the region where lignite is detected is named as Karapinar-Ayrancı lignite basin. 1.8 billion tons of coal resource was found by the drillings carried out by MTA in the lignite area with an exploration license belonging to EÜAŞ (2017). In his study, Salman (2010) determined that the coal is in the brown lignite class, and cannot be used for heating purposes due to its lower heating value and 20% ash content, but it can be a raw material for electricity generation in thermal power plants. Despite the abundance of lignite reserves in the region, there is no thermal power plant yet. It is thought that the construction of a thermal power plant in the region will be a great gain for the regional economy and for Turkiye in reducing foreign dependency on energy.

Coal production is a dangerous and dynamic complex system. Accident risk factors (methane gas, fire, flooding) affect and restrict coal production safety (Balovtsev et al., 2022; Bosikov et al., 2023; You et al., 2021). Besides the proliferation of sinkholes in the region made the Karapinar basin the most active karstic region. The fast emergence of sinkholes around Karapinar over the past 20 years presents a serious geological risk to the local populace, industry, and agriculture, as well as a threat to public health and safety in Karapinar (Dursun, 2022). Sinkholes are difficult to forecast because they can be caused by a number of processes, including the internal erosion of soil in the overburden and the downward dissolving of soluble materials. They are thus among the most hazardous geohazards in karst (Parise, 2019). Uncontrolled surface collapse also threatens the production and safety of mines, worsens the environment in mining areas, and changes hydrogeological and engineering geological conditions. Some paleo-sinkholes have great connectivity and permeability, making them ideal links between the coal seam and the karst aquifer. As a result, during the process of extracting coal, catastrophic water inrushes occur often, resulting in mine accidents, worker casualties, and environmental degradation (Gongyu & Wanfanf, 1999). There are many studies published in the literature about sinkholes and their environmental effects (Dougherty, 2005; Gao et al., 2013). There are also some studies on the effects of sinkholes on mining (Ścigała et al., 2023; Tzampoglou & Loupasakis, 2017).Karagüzel et al. (2020) obtained thematic maps for the Afşin—Elbistan coal basin, showing the spatial distribution of the parameters that determine sinkhole formation, using the GIS-based analysis method and AHP approach. De Bruyn and Bell (2001) reported a sinkhole occurrence during water pumping in a South African gold mine, leading to the tragic loss of 38 workers' lives. However, no study has been found in the literature on the selection of plant locations in regions containing sinkholes.

The site selection of thermal power plants is a strategic decision that has a significant impact on the economic operation of thermal power plants and the sustainable development of the region (Choudhary & Shankar, 2012). Site selection of a thermal power plant requires the consideration of many criteria together. Therefore, site selection is a multi-criteria decision-making problem (Ozer, 2005). These multi-criteria decision-making problems provide convenience to decision-makers in designing and solving complex problems involving many criteria or in evaluating possible alternative ways (Akdeniz et al., 2023; Feizizadeh et al., 2014). With the development of Geographic Information Systems, these methods have been used in applications such as urban and regional planning, land suitability mapping and site selection (Goldberg et al., 2023; Nasrollahi et al., 2023; Uyan & Yalpir, 2016). There are extensive reviews in the literature in terms of spatial investigation of the installation of facilities utilizing renewable energy sources; solar panels, wind turbines or wave energy (Azevêdo et al., 2017; Azizi et al., 2014; Eddine Boukelia & Mecibah, 2013). These studies have common spatial characteristics (power transmission line, slope, road, land use, etc.). However, compared to other studies, there are fewer coal-fired thermal power plant site selection studies in the literatüre.

Few studies focus on thermal power plant site selection using multi-criteria decision-making methods when examining the literature. Ramos et al. (2000) conducted a site selection study emphasizing financial risks in thermal power plant site selection in Sao Paulo, Brazil. Wang and Xin (2011) validated the model by analyzing how TOPSIS, which combines quantitative and qualitative indices, is applied in thermal power plant site selection. Kaliraj and Malar (2012) considered four primary criteria, land use, water availability, coal mine and environmental factors, and two secondary criteria, settlements and accessibility, by using remote sensing and GIS in an integrated way for thermal power plant site selection in Guajarat state of India. Choudhary and Shankar (2012) used fuzzy AHP and TOPSIS methods together to evaluate and select the most suitable sites for thermal power plants. Sambasivarao et al. (2014) proposed an expert system called Flex Expert System Shell, which analyzes the proposed site for the installation of thermal power plants according to the criteria set by the governing bodies and decides the suitability of the plant for operation. Xu et al. (2015) identified potential feasible sites for coal-fired power plant site selection using a GIS technique based on spatial analysis of several factors such as roads, power grids, fuel sources, etc. Siefi et al. (2017) emphasized that economic growth and development activities in the Kerman Province of Iran increase the annual demand for energy supply, thus new power plants are required, and they identified the most suitable locations available by using environmental and socio-economic criteria supported by weighted linear combinations.

In this study, considering the presence of rich lignite reserves discovered in the Karapinar region and the lack of a thermal power plant that can utilize this lignite reserve and generate electricity in the region, a site was selected for the thermal power plant by considering a total of 12 criteria under 3 main criteria by using GIS and MCDM methods to provide ideas for various studies to be carried out in the future.

During the site selection, the Best–Worst Method (BWM) and Technique for Order Preference by Similarity to the Ideal Solution (TOPSIS) method were used together for criteria weighting and determination of the most suitable location. With such a study, creates a methodology for managers on how they should proceed in areas with great risk (such as the Karapınar region, which has the highest number of sinkholes in Turkey).

Materials and methodology

Study area and workflow

The study area is the Karapinar district of Konya province in Turkiye (Fig. 1). In terms of its coordinates, it is located in the WGS-84 datum between 37°27′35.07″ and 38°0′25.39″ North latitudes and 33°7′12.63″ and 33°58′29.15″ East longitudes and has an approximate surface area of 2,747 km^2. The distance between the district and Konya city center is 102 km and its average height above sea level is 1.026 m (Karapinar District Report, 2019). The population of the district was calculated as 50,055 people in 2021 (URL-2). As of the age of 20, the population is decreasing day by day in the district and one of the main reasons for this may be employment (Karapinar District Report, 2019). It is thought that the installation of a thermal power plant will increase employment opportunities and will also stop the migration of the young population.

Karapinar is located in a region known as the “Sinkhole Plateau”, the formation of which has been going on for millions of years. Today, sinkhole formation has accelerated in that region, especially due to global warming and excessive use of groundwater for agricultural irrigation (Fig. 2a). It is observed that the sinkholes in the Karapinar region were far away from residential areas in the past. Still, in recent years they have started to form along highways and near residential areas (Fig. 2b). There are more than 550 sinkholes in the Karapinar region. In terms of axis length, Sivritepe with 900 m ranks first among the basement rocks and Potur with 820 m ranks first among the younger rocks. The “deepest” sinkhole is Cirali with 134 m (Fig. 2c).

Within the scope of the study, a total of 12 sub-criteria including the sinkholes were determined under 3 main criteria. Maps were obtained with various analyses using Arc-GIS 10.5.1 version, the GIS software, to meet the criteria. Each criterion was weighted by the decision makers using BWM to obtain the suitability map. The suitability map for thermal power plant site selection was created with the weighted combination method in Arc-GIS software by using the criteria weights (Fig. 3). In the suitability map, 4 candidate sites with the highest suitability value were determined. The TOPSIS method was used to determine the most suitable location among the candidate sites (Fig. 3).

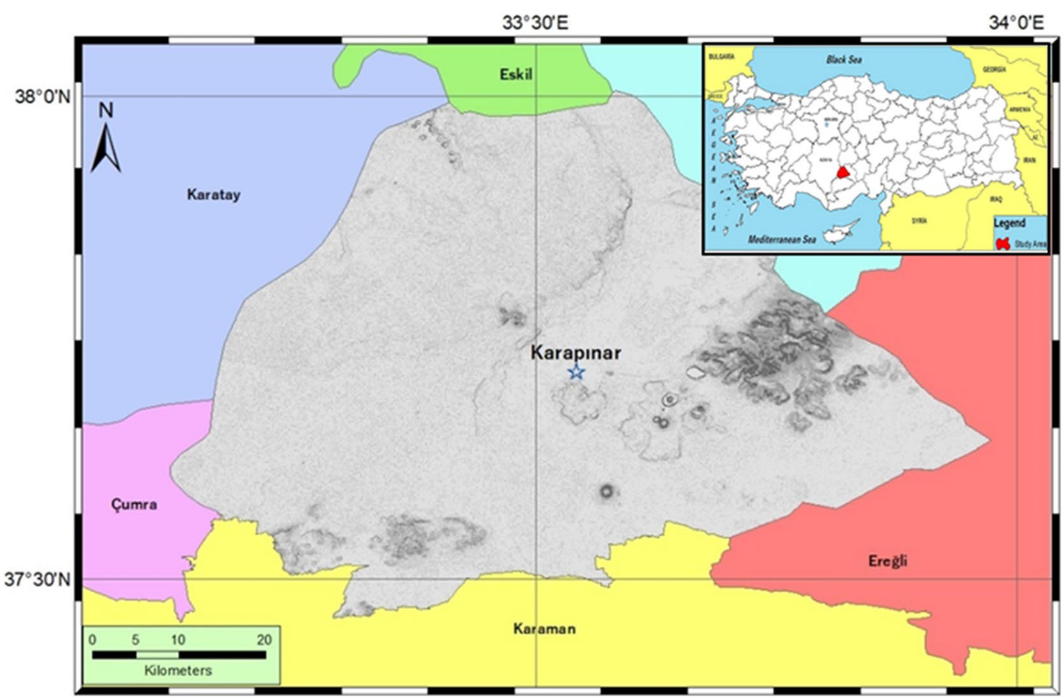

Fig. 1 Location of the study area

Fig. 2 Example of sinkhole formations in the Karapinar region (**a**) newly formed sinkhole (**b**) near the residential area (**c**) old-formed Cirali sinkhole

Criteria affecting thermal power plant installation

The aim of the study is to determine the most suitable site for thermal power plant installation in the Karapinar region. For this purpose, the criteria and restricted areas to be evaluated in selecting the installation site were determined and a data set was created (Table 1). The criteria used in the study were determined through expert opinions (Environmental and Surveying Engineers, Mining and Geology) and literature review. In the site selection studies of thermal power plants, Choudhary & Shankar, 2012 utilized geological and soil type, land use, and electricity transmission line accessibility criteria; Xu et al., 2015 employed fault lines and archaeological site criteria; Siefi and colleagues considered slope, roads, groundwater sources, and settlement criteria; Akar et al., 2023 incorporated protected areas and forest criteria into their research. These criteria have been included in the study. In addition, the sinkhole formations, which are a unique feature of the study area, were added to the study as a sinkhole density criterion by the expert team, as they constitute the originality of the study.

Different sources were used in the creation of spatial map bases, which are effective in thermal power plant installation. Many of these sources are open-source databases. Other map bases were arranged by digitizing the existing maps. In the created data set, the criteria were grouped under 3 main criteria: geological, economic and environmental. Geological main criterion include distance to faults, earthquake epicenter and magnitude, sinkhole density, slope, lithology, groundwater level. Economic main criterion include proximity to roads, proximity to transmission line and distance to residential area. Environmental main criteria include land use, distance to protected areas and distance to forests (Table 1). In total, 12 criteria were evaluated in the study. Each criterion was evaluated according to the appropriate spatial analysis method and given a suitability degree between [1–4] to be used in creating the integrated suitability map (Table 1). After determining the criteria, Arc-GIS, a GIS software, was used for reclassification through analyzing and scoring the criteria (Reclassify).

Geological (C1) criteria

C11 (Distance to Active Faults) In case of earthquake hazard, the structures on active faults are directly exposed to the most impact. In case of damage to the thermal power plant structure, serious pollution and material damages occur. In this study, it is desirable for the thermal power plant installation site to be located at a significant distance from active faults. For this reason, the active fault lines in the study area were digitized from the Mineral Research and Exploration (MTA) active fault lines Karaman (NJ 36–11) map (URL-3).

C12 (Earthquake Epicenter and Magnitude) Seismic activity poses a threat to power plants. Therefore, it's crucial to consider the epicenter and magnitude of earthquakes when selecting plant locations, aiming to avoid these regions. The data on the coordinates and magnitude values of the seismic activities that occurred in and around the study area in the last year were obtained from the Kandilli Observatory and Earthquake Research Institute (URL-4). The epicenters of the earthquakes were added to the Arc-GIS environment as point data with the help of their coordinates.

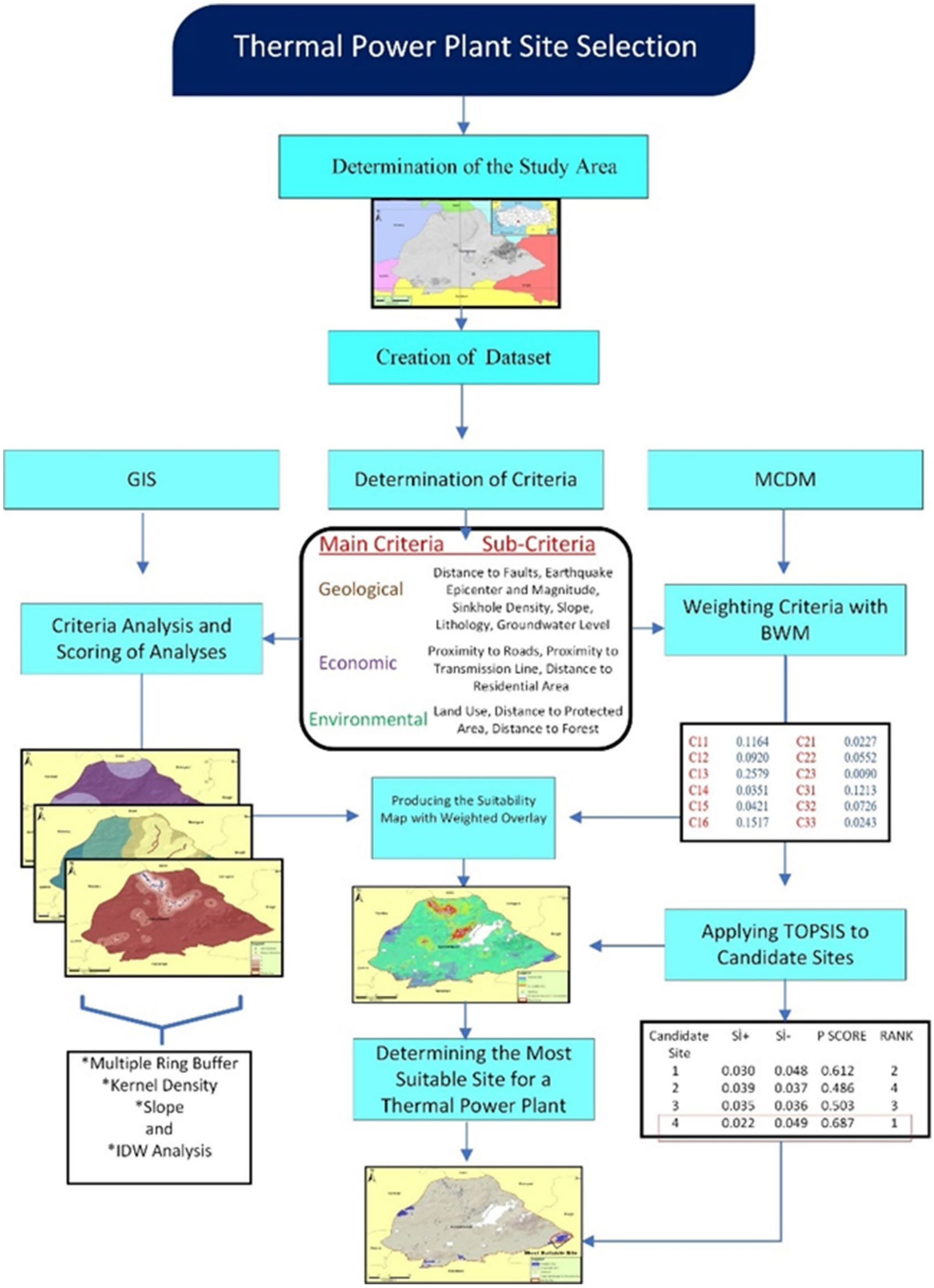

Fig. 3 Workflow diagram

Table 1 Main criteria, sub criteria and code name

Code	Criteria	Sub-Code	Sub-Criteria	Data Set References	Value	Point
C1	Geological	C11	Distance to Faults (km)	MTA	0–5 6–10 11–25 26–50	1 2 3 4
		C12	Earthquake Epicenter and Magnitude (M)	Kandilli Observatory and Earthquake Research Institute	0–1 1,1–2 2,1–3 3,1 >	4 3 2 1
		C13	Sinkhole Density	Orhan et al., 2020	0–0,9 1–1,9 2–3,9 4 >	4 3 2 1
		C14	Slope (%)	NASA-USGS	0–2 2,1–8 8,1–16 16,1–46,3	4 3 2 1
		C15	Lithology	MTA	Undifferentiated Quaternary Andesite Continental Clastic Rocks Marble	4 3 2 1
		C16	Groundwater level (m)	DSİ	12–115 116–134 135–171 172–280	4 3 2 1
C2	Economic	C21	Proximity to Roads (m)	Open Street Map	0–1000 1001–2500 2501–5000 5001–10000	4 3 2 1
		C22	Proximity to Transmission Line (m)	Open Street Map	0–500 501–1000 1001–5000 5001 >	4 3 2 1
		C23	Distance to Residential Area (km)	CORINE	0–1 2–5 6–15 16–25	1 2 4 3
C3	Environmental	C31	Land Use	CORINE	Open Land Non-Irrigated Land Irrigated Land Pasture	4 3 2 1
		C32	Distance to Protected Area (km)	Minister of Environment, Urbanization and Climate Change	0–5 6–10 11–15 16–25	1 2 3 4
		C33	Distance to Forest (km)	CORINE	0–5 6–15 16–25 26–50	1 2 3 4

C13 (Sinkhole Density) The formation of sinkholes is directly or indirectly related to geological, geomorphological, and hydrogeological factors (Ozdemir, 2016). In general, the carbonic acid formed by the reaction of groundwater with carbon dioxide dissolves rocks over time, creating cavities and cave systems under the ground, and over time, sinkholes occur due to the collapse of the structure on the ceiling (Orhan, 2018). Structures (housing, industry, commerce, etc.) cannot be built on or around sinkholes. Given the high density of sinkholes in the study area, it is a crucial criterion to consider when selecting a site in this region. Accordingly, it was requested that the site of the plant should be far away from the parts where the density of sinkholes is high. In addition, old sinkholes are often the best predictors of new sinkholes, especially where they show a clustered distribution (Gutierrez et al., 2016). Therefore the sinkholes in the study area were digitized from the sinkhole map in an existing study (Orhan et al., 2020).

C14 (Slope (%)) The slope reflects the flatness of the land in the study area and a high slope means a more difficult terrain for the installation of power plants (Xu et al., 2020). A high slope also increases construction costs as it requires excavation and filling work. The digital elevation model required for the slope analysis of the study area was obtained from NASA-USGS (URL-5).

C15 (Lithology) The properties of the ground on which a structure will be built are crucial for the stability of the structure, particularly for power plants. The geological structure of the study area was obtained through digitization from the website of MTA (URL-6). In terms of lithology, the study area is mainly classified into four classes: unseparated Quaternary, andesite, terrestrial clasts and marble.

C16 (Underground Water Level) Thermal power plants require cooling water to operate, and, given the scarcity of surface waters in the study area, the presence and depth of groundwater were carefully considered. Water well data of the study area and its immediate surroundings were obtained from State Hydraulic Works. These water well locations were digitized as point data in the ArcGIS environment, with the groundwater levels of each well included as attribute data.

Economic (C2) criteria

C21 (Proximity to roads) To minimize transportation costs during construction and the subsequent transportation of raw materials, proximity to existing roads was considered. Road data for the study area and its surroundings were sourced from OpenStreetMap open data (URL-7).

C22 (Proximity to Power Transmission Line) Since the use of existing power transmission lines would be more economical for the operation, the criterion of proximity to power transmission lines was considered. Power transmission line data for the study area and its immediate surroundings were downloaded via Open Street Map (URL-7).

C23 (Proximity to Residential Areas) To balance the environmental impact and the need for manpower, thermal power plants should neither be too close to residential areas nor too far from them, both during construction and operation. The land use map depicting residential areas in and around the study area was acquired from CORINE (URL-8).

Environmental (C3) criteria

C31 (Land Use) The thermal power plant's location should avoid agricultural lands and the presence of fertile soil in its immediate surroundings. To achieve this, the land use of the study area was categorized into four groups and graded from the most desirable to the least desirable: open area, dry land, irrigated land, and pastureland. The land use map for the study area was sourced from CORINE (URL-8).

C32 (Distance to Protected Areas) Due to the impacts of the thermal power plant on the environment, it should be far away from protected natural and artificial areas. Protected areas, natural areas and nature parks in and around the study area were obtained from the Protected Areas Management System developed by the Ministry of Environment, Urbanization and Climate Change (URL-9).

C33 (Distance from Forests) Thermal power plants should be far away from green areas due to the negative effects they may have on the natural environment. Data on the forests in the study area and its vicinity were obtained from the CORINE land use map (URL-8).

Geographic Information Systems (GIS)

GIS technology has emerged as a new information processing technology mainly in the acquisition, management and analysis of spatial data since the 1980s (Erden & Coskun, 2011). GIS, which performs the functions of collecting, storing, analyzing and presenting graphical and non-graphical data obtained by location-based operations, is an information system that provides meaningful outputs by using different decision-making methods in solving environmental issues (Guler, 2016; Yomralioglu, 2000). Considering these features, Arc-GIS 10.5.1, a GIS software, was used in the study to analyze and reclassify the criteria for thermal power plant site selection and to create the suitability map by weighted combination.

After the study area and data sets were prepared for the selection of the thermal power plant installation site, Slope (C14), distance/buffer zone (Multiple Ring Buffer) (C11, C21, C22, C23, C32 and C33), density (Kernel Density) (C12 and C13), IDW (C16) analyses were applied to the data sets transferred to Arc-GIS environment, a GIS software. The maps obtained after the analysis were converted to raster format and reclassified by scoring between [1–4] (Table 1). All classified maps in raster format and their weights were combined with the raster calculator to create a suitability map of the thermal power plant site. For the most suitable site, BWM methods were used to determine the weights of the criteria and TOPSIS methods were used to evaluate the candidate sites.

Multi-Criteria Decision Making (MCDM) Methods

Spatial decision problems include a large set of alternatives to be evaluated by a large number of decision-makers and incompatible evaluation criteria that conflict with each other within this set. Decision analysis techniques are used to solve spatial decision problems. MCDM methods, among the decision analysis techniques, provide many techniques and procedures for structuring decision problems and designing, evaluating and prioritizing alternative decisions (Malczewski, 2006). GIS technologies, which have developed with the developments in technology, are now used in spatial decision-making problems together with MCDM methods.

They used the AHP method, which is the most preferred method in the literature among the MCDM methods, for selecting the locations of offshore wind power plants with over 17 criteria and for spatial suitability of wind energy with 9 criteria in the studies (Caceoglu et al., 2022; Villacreses et al., 2017). Wang & Xin, in 2011, analyzed the pros and cons of different plans regarding the location of the thermal power plant and reached the most appropriate solution using the TOPSIS method. In addition, in the ANP method, which is preferred for the installation site selection problems of geothermal and wind power plants, which are renewable energy sources, a network model was created by first determining the main criteria, sub-criteria and alternatives for each facility (Taraf & Yazgan, 2018). Therefore, multi-criteria decision making methods are used to solve the site selection problem. In the literature, it is seen that BWM and TOPSIS methods are not used together in the thermal power plant site selection. Therefore, in this study, BWM and TOPSIS, which are among the MCDM methods, were used together to determine the thermal power plant installation site for weighting the criteria and site selection to contribute to the literature.

Best–Worst Method and TOPSIS

BWM (best–worst method) is a multi-criteria decision-making method called pairwise comparison and developed by Dr. Jafar Rezaei. The method aims to derive the weights of the criteria based on a pairwise comparison of the best and worst criteria. A five-step procedure was used to derive the weights, and a consistency ratio was developed to check the reliability of the final results (Rezaei, 2015). BWM was used by decision-makers to determine the weights of the criteria within the scope of the study and to create a suitability map for thermal power plant site selection.

TOPSIS, developed by Hwang and Yoon in 1981 (Behzadian et al., 2012). The method was later developed by Chen and Hwang in 1992 and the main purpose of the method is to ensure that the selected alternative is closest to the ideal solution and farthest from the negative ideal solution (Alshamrani et al., 2023; Karaatli et al., 2015). TOPSIS, which allows the candidate regions to be ranked according to their degree of suitability, was applied to determine the most suitable region for the thermal power plant installation site among the candidate regions selected from the suitability map. The processing steps of the BWM and TOPSIS methods are given in Table 2.

Integration of BWM and GIS via weighted overlay analysis

The map data set to be used in the study requires a series of processes to be used in GIS. These processes allow integration into the BWM model as they are placed in the workflow diagram (Fig. 3). In order to equalize the numerical scales of the criteria, it is necessary to standardize the data by either reclassification to be able to evaluate them in the same analysis. In this study, the resultant maps were created by integrating the criteria weighting coefficients obtained in the BWM and the datasets prepared in the GIS environment via the weighted overlay analysis with the formula given in Eq. 1.

$$TPPI = \sum_{j=1}^{n} W_j x_{ij} \tag{1}$$

where TPPI: Thermal Power Plant Index; n: number of criteria; j: criteria no j = 1 to n; Wj: BWM weight of jth criteria; x: pixel values of reclassed maps from GIS; i: pixel no (Doorga et al., 2019; Malczewski, 1999).

Findings

Considering the borders of Karapinar district, the application results of the selection of the thermal power plant installation site, which has the potential to meet the energy needs of the region and reduce the dependence on foreign energy throughout the country, were evaluated. As a result of the suitability map created by using the weights of the criteria presented in Table 1 determined by the BWM, 4 candidate regions were identified. Figures 4, 5,and 6 show the maps of the analysis and reclassification scoring applied to the sub-criteria under the main criteria of geological, economic and environmental, respectively.

When the maps of the results of the analysis made under the main criterion of geological are examined, it is seen in the map of active fault lines (Fig. 4a) that the active fault lines are located in the north-east of the study area and come towards the district center. When the epicenter and magnitude of earthquakes are analyzed, it is understood that the earthquakes that occurred in the last year are concentrated in the regions close to the southeast of the study area (Fig. 4b). It is observed that most of the sinkholes in the region are concentrated from the center of the study area to the north (Fig. 4c) and there are no high slope values except for the mountainous region to the east of Karapinar (Fig. 4d). When the lithology map (Fig. 4e) is examined, it is seen that the lithology considered suitable for site selection extends mostly from the southern half of the study area to the east. It is understood from the analysis that the areas of Karapinar region where the groundwater level is close to the surface extend from the south to the west of the study area (Fig. 4f). As a result of the analysis, classified maps were obtained for the six sub-criteria included in the geological main criteria.

According to the maps of the results of the analysis made under the economic main criterion, it was revealed that there is access to the road network in the study area in general, and that especially the center and southwestern parts of the study area have access to the dense transportation network (Fig. 5a). It is seen that there is a power transmission line passing from the west to the east and southeast of the study area and a main line in the southern part (Fig. 5b). When the residential areas (Fig. 5c) are examined, it is understood from the analysis maps that there are dense residential areas in the center and southwest of the study area.

When the maps of the results of the analysis made under the environmental main criterion of the Karapinar region are examined, it is seen that the land structure considered suitable for site selection is concentrated in the southeast of the study area (Fig. 6a), and there are protected areas on a large scale on the borders of the neighboring regions located to the northeast and south and west of the center of the study area (Fig. 6b), and there are forests within the neighboring district located to the east and south-west of the study region (Fig. 6c).

Restricted Areas Areas where the thermal power plant cannot be built are considered as restricted areas. Due to the risk of sudden formation, especially the sinkholes included in the study, residential areas, protected areas, archaeological excavation areas, swamps, lakes and forests were determined as restrictive areas. Solar panels and swamps in the part from the center of the study area to the northeast, and

Table 2 BWM and TOPSIS process steps

STEP	BWM		TOPSIS	
1	The decision criteria to be used in the study are determined.($c_n = Criteria$)	$(c_1, c_2, \dots, c_n)$	The alternatives to be ranked and the criteria against which these alternatives will be compared are determined	
2	Decision makers determine the best (most important) and worst (least important) criteria		A decision matrix is created. The rows of the matrix contain the alternatives to be ranked and the columns contain the evaluation criteria to be used in decision making. (a_{ij}= i shows the score of the candidate region according to criterion j)	$A_{ij}= \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$
3	The priority of the best criterion over all other criteria is determined by using a number between 1 and 9. As a result, the best-to-other vector is obtained. Here, a_{Bj} indicates the priority of the best criterion B over criterion J	$A_B = \left(a_{B1}, a_{B2}, \dots, a_{Bn}\right)$	The decision matrix is normalized. (r_{ij}= Normalized matrix values, v_{ij}= Weighted normalized matrix values)	$r_{ij} = \frac{w_{ij}}{\sqrt{\sum_{i=1}^{j} w_{ij}^2}}$ v_{ij}=$w_i * r_{ij}$ $(j=, 1, 2, 3, \dots, J\ i=1, 2, 3, \dots, n)$
4	The priority of all criteria over the worst criterion is determined using a number between 1 and 9. As a result, the vector with the worst of the other criteria is obtained. Here, a_{jW} indicates the priority of criterion J over the worst criterion	$A_W = \left(a_{1W}, a_{2W}, \dots, a_{nW}\right)^T$	Positive ideal solution and negative ideal solution are determined	$A^* = \left\{v_1^*, v_2^*, \dots, v_n^*\right\}$ maximumvalues $A^- = \left\{v_1^-, v_2^-, \dots, v_n^-\right\}$ minimum values
5	The criteria have optimum weights $\left(W_1^*, W_2^*, \dots, W_n^*\right)$. The optimal weight for the criteria is W_B/W_j =a_{bj} and W_B/W_j =a_{jW} for each pair of W_B/W_j and W_j/W_W. It is necessary to find a solution where the maximum absolute differences are minimized	$\left\lvert \frac{W_B}{W_j} - a_{Bj} \right\rvert \leq \xi, \forall j$ $\left\lvert \frac{W_j}{W_W} - a_{jW} \right\rvert \leq \xi, \forall$ $\sum_{W_j} = 1 W_j \geq 0, \forall j$	The distance of each alternative from the positive ideal solution and the negative ideal solution is calculated. (d_i^*= distance from the positive ideal solution, d_i^- = distance from the negative ideal solution)	$d_i^* = \sqrt{\sum_{j=1}^{n}\left(v_{ij} - v_j^*\right)^2}$ $d_i^- = \sqrt{\sum_{j=1}^{n}\left(v_{ij} - v_j^-\right)^2}$,
6	The optimum weights of the criteria and ξ are obtained with the solution of the model. After obtaining the weights, the consistency ratio is calculated using the consistency index values in Table 2 as the final step (Rezaei, 2015)	Consistency Ratio $= \frac{\xi}{ConsistencyIndex}$	The proximity coefficient (CC_i) of each alternative is calculated. By comparing CC_i values, the order of preference of the alternatives is determined (Karaatlı et al., 2015; Yoon & Hwang, 1995)	$CC_i = \frac{d_i^-}{d_i^* + d_i^-}$ $i=1, 2, \dots, J$ $0 \leq CC_i \leq 1$

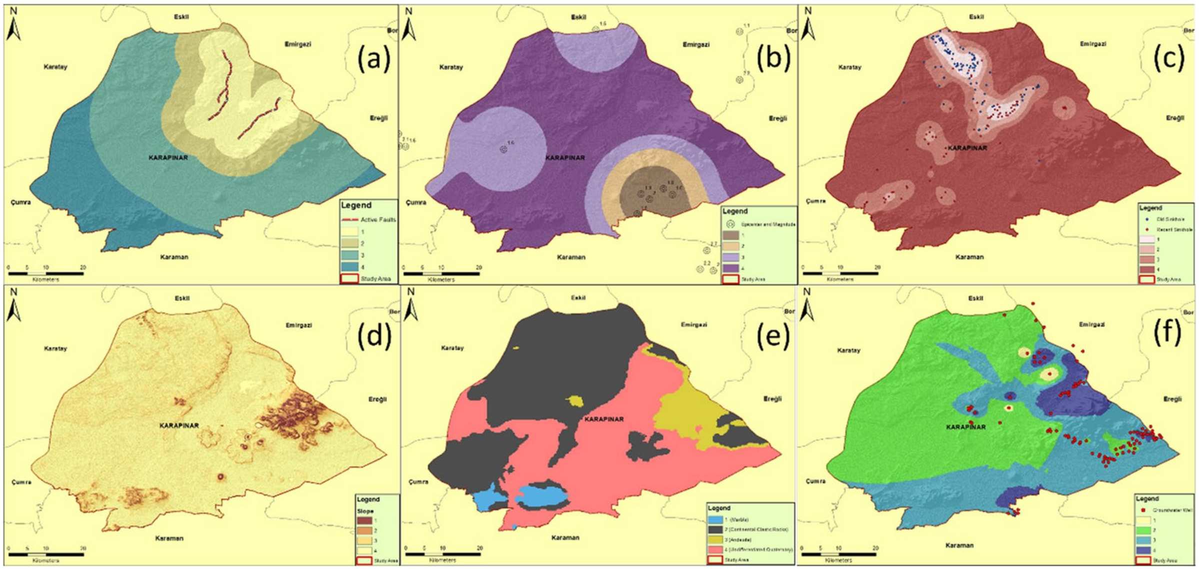

Fig. 4 **a** Distance to Faults, **b** Earthquake Epicenter and Magnitude, **c** Sinkhole Density, **d** Slope, **e** Lithology, **f** Groundwater Level

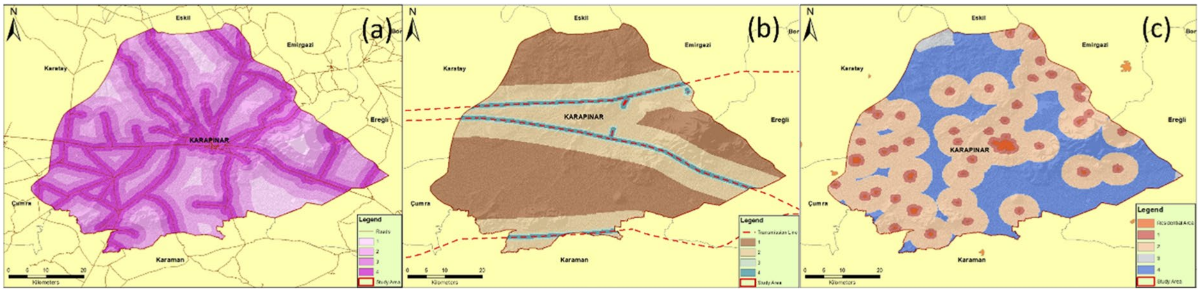

Fig. 5 **a** Proximity to Roads, **b** Proximity to Transmission Line, **c** Proximity to Residential Area

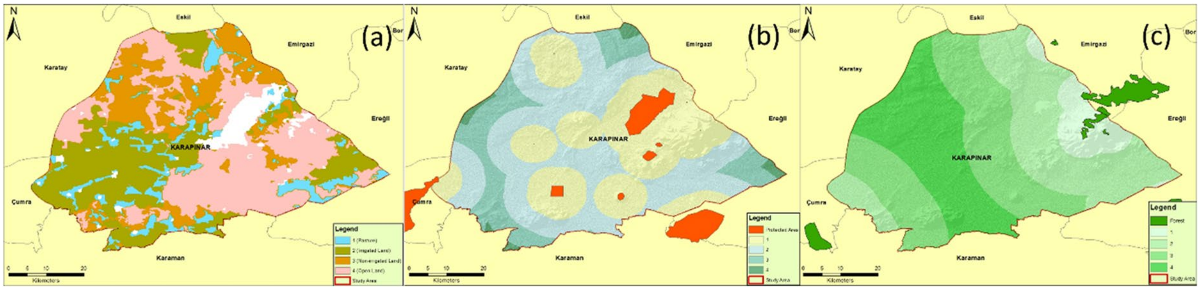

Fig. 6 **a** Land Use, **b** Distance to Protected Area, **c** Distance to Forest

especially sinkhole formations in the areas extending from the center to the north are the areas that should never be included in the thermal power plant installation (Fig. 7). Restricted areas constitute approximately 7% of the entire study area.

Weighting of criteria using the Best–Worst Method

In this part of the study, the final weights of a total of 12 sub-criteria under 3 main criteria were determined with BWM by taking the opinions of 10 decision makers (consisting of Mining, Geology, Mechanical, Environmental and Surveying Engineers). The criteria covered in the study and code names to be used in the BWM are shown in Table 1. Decision makers evaluated 3 main criteria and a total of 12 sub-criteria related to these main criteria according to BWM. Firstly, the decision makers determined the criteria groups as the best and the worst within themselves. Then, using the evaluation scores according to Saaty (1980) from the best to the others and from the others to the worst, the degree of importance was determined by following the procedure steps shown in Table 2. Table 3 shows the decision makers' scores for the main criteria.

After determining the best and worst criteria and their importance levels for the main criteria, a model was established according to the equations in Table 2 and then analyzed. As a result of the analysis, the results of the criteria weights, ξ, consistency ratios and average values of the decision makers were calculated for the main criteria (Table 4).

The same procedures were performed for the sub-criteria under the main criteria C1, C2 and C3. In order to find the final weights of the 12 sub-criteria, the average weight of each sub-criterion was multiplied by the average weight of the main criteria to which it belongs. As a result, the weights to be used in the Weighted Overlay process were calculated in Arc-GIS. Table 5 shows the weights, ξ, consistency ratios of the 3 main criteria and sub-criteria and the final weights of the sub-criteria.

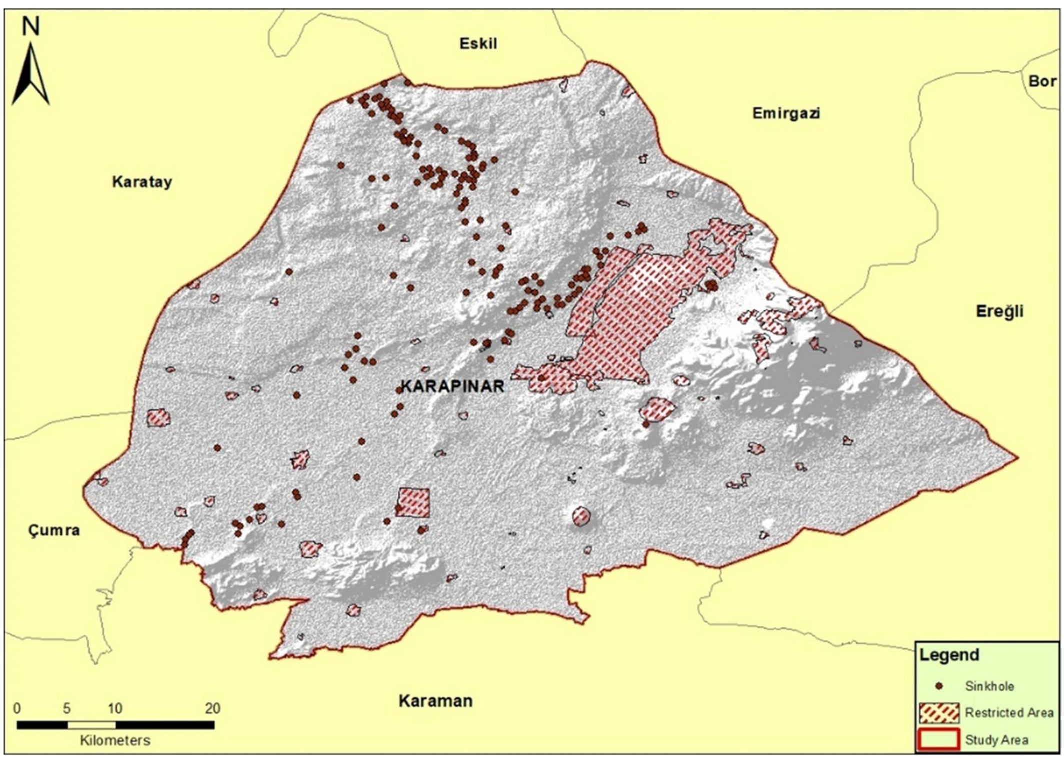

Fig. 7 Restricted Areas in the Study Area

Table 3 Evaluation of main criteria of decision makers

Decision maker 1: Determining The Best and Worst Criteria			
Best Criterion: C1		Worst Criterion: C2	
Determining The Preference of The Best Criterion Over All The Other Criteria			
	C1	C2	C3
Best Criterion (C1)	1	7	4
Determining The Preference of All The Criteria Over The Worst Criterion			
	C1	C2	C3
Worst Criterion (C2)	7	1	5
Decision Maker 2: Determining The Best and Worst Criteria			
Best Criterion: C1		Worst Criterion: C2	
Determining The Preference of The Best Criterion Over All The Other Criteria			
	C1	C2	C3
Best Criterion (C1)	1	7	3
Determining The Preference of All The Criteria Over The Worst Criterion			
	C1	C2	C3
Worst Criterion (C2)	5	1	2
Decision Maker 3: Determining The Best and Worst Criteria			
Best Criterion: C1		Worst Criterion: C2	
Determining The Preference of The Best Criterion Over All The Other Criteria			
	C1	C2	C3
Best Criterion (C1)	1	7	5
Determining The Preference of All The Criteria Over The Worst Criterion			
	C1	C2	C3
Worst Criterion (C2)	7	1	5

Table 4 Main criteria weights, consistency ratios and average values of decision makers

	W_{C1}	W_{C2}	W_{C3}	ξ	CR
DM-1	0.7051	0.0769	0.2179	0.1667	0.0447
DM-2	0.6429	0.1071	0.2500	0.1071	0.0287
DM-3	0.7363	0.0769	0.1868	0.1978	0.0530
Average Values	0.6947	0.0870	0.2183	0.1572	0.0421

Generating suitability map using GIS

The final weight values of the criteria were used in the Weighted Overlay process in the Arc-GIS environment to produce a suitability map (Fig. 8). On the map, areas in white represent restricted areas. The suitability map is shown as a color scale from the most unsuitable area to the most suitable area. Accordingly, the colors changing from red to blue represent the regions ranging from the most unsuitable area to the most suitable area. When the red-colored regions are examined, it is seen that the sinkhole formation is especially intense and concentrated in the regions where the fault lines are located. Considering the blue colors, the areas that can be shown as suitable cover approximately 10% of the entire study area. Considering the areas that can be described as the most suitable areas, it is observed that they are concentrated in the western, southern and eastern ends of the study area. From these areas, an area of approximately 46 km^2 was selected a candidate regions.The selected 4 candidate regions constitute approximately 2% of the study area.

Candidate region ranking using TOPSIS Method

In the last stage of the study, the TOPSIS method was applied based on 4 candidate regions with high suitability ratios obtained from the suitability map and alternative sites were ranked for the location of the thermal power plant (Fig. 9). Looking at the candidate regions in order, it is seen that Candidate Region 1 is in the west of the study region and covers an area of 17.6 km^2. Candidate Region 1, which is 30 km away from Karapinar city center, also includes two residential areas with a total area of 620,000 m^2, which are considered as restricted areas. Candidate Region 2 is in the south of the study area, close to the Karaman border and covers an area of 4.8 km^2. Candidate Region 2, which is 31 km away from Karapinar city center, is also on the border of a 2 km^2 residential area, which is considered a restricted area. Candidate Region 3 has an area of 4.5 km^2 and is in the south of the study area and 12 km north-east of Candidate Region 2. Candidate Region 3 is also on the border of a 590,000 m^2 residential area and 22 km away from Karapinar city center. Candidate Region 4 is at the easternmost end of the study area and covers an area of 19 km^2. This region is 34 km from Karapınar city center and the farthest region from sinkhole density among the other candidate regions (Fig. 9).

In order to apply the TOPSIS method, first the candidate regions to be ranked and the criteria to compare these regions were arranged as a decision matrix. The score values of the criteria corresponding to the candidate region, as indicated in Table 1 were written

Table 5 Final weights of sub-criteria

Main Criteria	Weights	CR	Sub-Criteria	Weights	CR	Final-Weights
C1	0.6947	0.0421	C11	0.1676	0.0221	0.1164
			C12	0.1324		0.0920
			C13	0.3712		0.2579
			C14	0.0506		0.0351
			C15	0.0607		0.0421
			C16	0.2176		0.1517
C2	0.0870		C21	0.2615	0.0340	0.0227
			C22	0.6348		0.0552
			C23	0.1037		0.0090
C3	0.2183		C31	0.5557	0.0349	0.1213
			C32	0.3328		0.0726
			C33	0.1115		0.0243

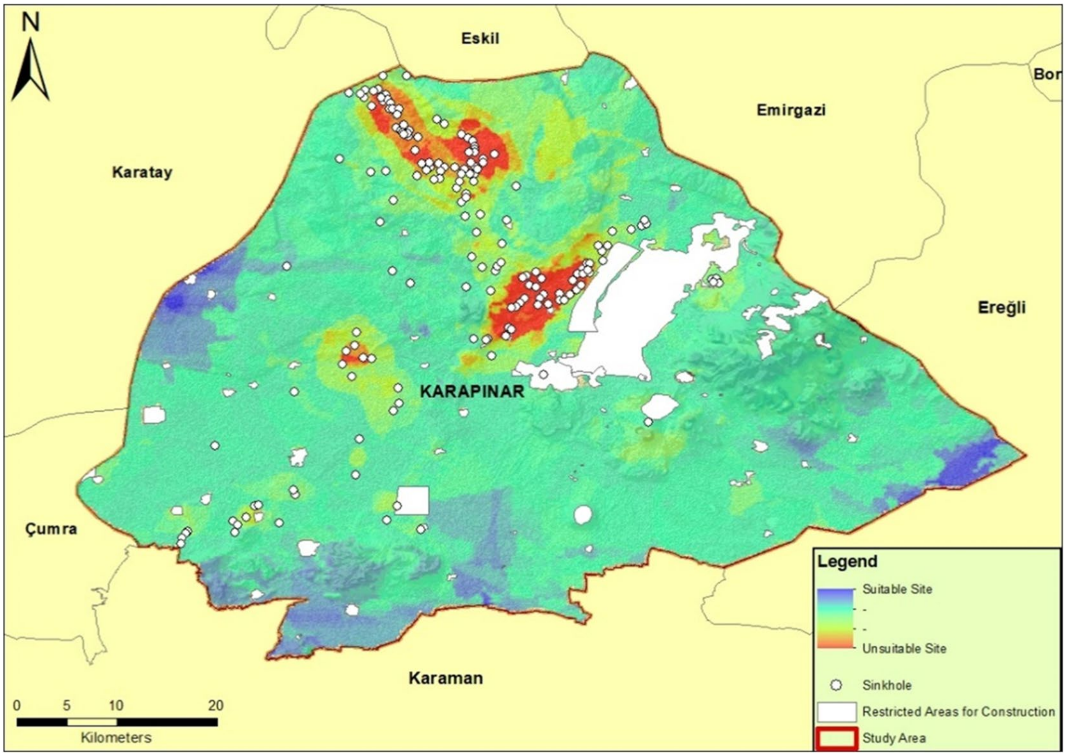

Fig. 8 Suitability map for thermal power plants

in the cells of the decision matrix. Then, the decision matrix was normalized as shown in Table 2. The weighted normalized matrix was obtained by multiplying the weights of each criterion by the cell values corresponding to its own column in the normalized matrix. Positive and ideal solution values were calculated with the help of the equations in Table 2 (Table 6).

The distance of each alternative from the positive and negative ideal solutions were calculated using the equations in Table 2. Then the proximity coefficient was calculated and named as the P Score. Starting

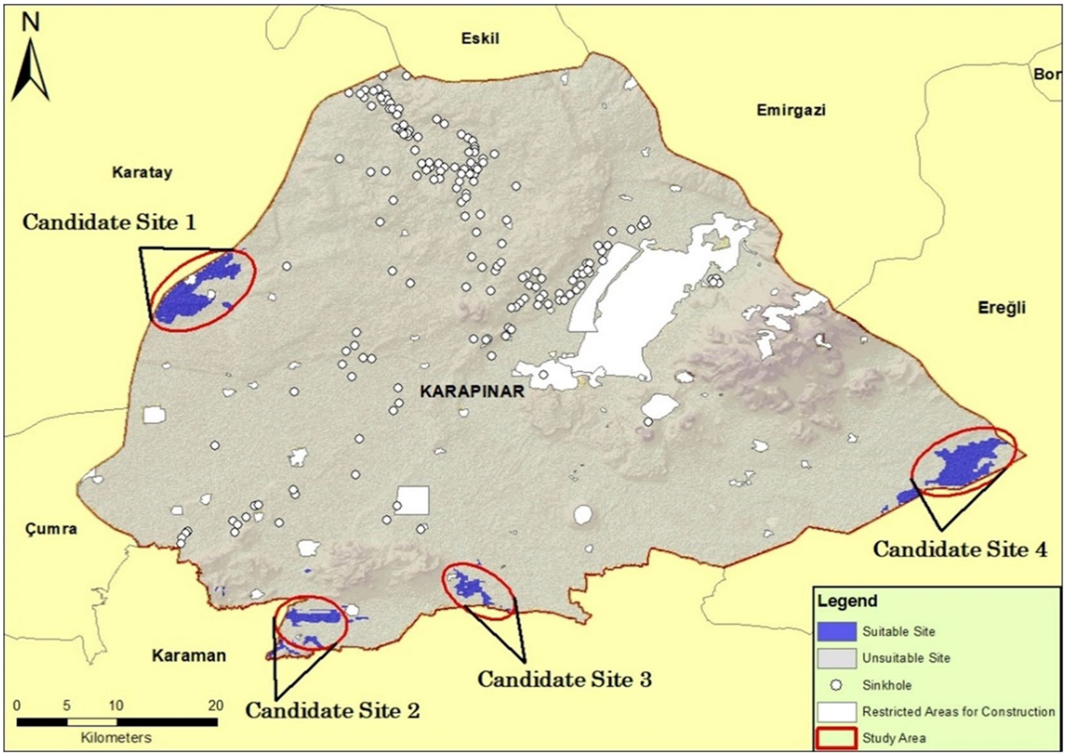

Fig. 9 Map of candidate sites

with the one with the highest P Score coefficient, the most suitable locations for the thermal power plant installation site were listed (Table 7). After the application of the TOPSIS method, it was determined that the most suitable area for the thermal power plant installation site in Karapinar district is the candidate region 4. As an alternative to Candidate Region 4, Candidate Region 1, which ranks second in terms of the P Score coefficient, can also be considered for site selection.

Candidate Region 4 is the one with the largest area among the regions. The region, which is far from the center and north of the study area, where the sinkhole density is highest, is neither too close nor too far from the residential areas. Therefore, it is in an advantageous location in terms of providing a labor force and having a low pollution impact on residential areas. In addition, the region on the border with Eregli has transportation facilities that can contribute to providing labor force and reducing construction costs. The region, which is far away from faults and earthquake-prone areas, is also advantageous in terms of the safety of the plant. In addition, the fact that the plant will be built in a suitable location in terms of lithology indicates that the plant will be safe from various seismic activities. In general, the region, which is also far from the natural protection areas, can be affected the least in adverse situations that may occur. The fact that the power transmission line passes very close to the region is also economically advantageous for the plant. In addition, considering the most suitable region for site selection, the fact that the plant will be installed in an open area according to the land use map shows that it will be in a location that will not prevent agricultural activities. Considering the geological, economic and environmental criteria within the scope of the study area, Candidate Region 4, which was selected as the most suitable region among the 4 regions as a result of TOPSIS, is seen to be in the most advantageous location compared to other regions in terms of all criteria. Considering Candidate Region 1, which is determined as an alternative, the region on the border of Karatay district is the candidate region which is in the westernmost part of the

Table 6 Decision matrix, normalized matrix, weighted normalized matrix and positive and negative ideal solutions

	Candidate Sites	C11	C12	C13	C14	C15	C16	C21	C22	C23	C31	C32	C33
Decision Matrix	Site 1	4	4	4	3	2	2	4	4	2	4	4	4
	Site 2	4	4	4	4	4	3	2	4	2	2	3	3
	Site 3	4	4	4	3	4	3	3	2	2	3	2	4
	Site 4	4	4	4	4	4	3	4	2	4	4	3	2
Normalized Matrix	Site 1	0.50	0.50	0.50	0.42	0.28	0.36	0.60	0.63	0.38	0.60	0.65	0.60
	Site 2	0.50	0.50	0.50	0.57	0.55	0.54	0.30	0.63	0.38	0.30	0.49	0.45
	Site 3	0.50	0.50	0.50	0.42	0.55	0.54	0.45	0.32	0.38	0.45	0.32	0.60
	Site 4	0.50	0.50	0.50	0.57	0.55	0.54	0.60	0.32	0.75	0.60	0.49	0.30
Weighted Normalized mMMatrix	Site 1	0.058	0.045	0.129	0.015	0.012	0.054	0.013	0.035	0.003	0.072	0.047	0.014
	Site 2	0.058	0.046	0.129	0.020	0.023	0.081	0.007	0.035	0.003	0.036	0.035	0.011
	Site 3	0.058	0.046	0.129	0.015	0.023	0.081	0.010	0.017	0.003	0.054	0.023	0.014
	Site 4	0.058	0.046	0.129	0.020	0.023	0.081	0.013	0.017	0.007	0.072	0.035	0.007
Positive and Negative Ideal Solutions	V +	0.058	0.046	0.129	0.020	0.023	0.081	0.013	0.035	0.007	0.072	0.047	0.014
	V-	0.058	0.046	0.129	0.015	0.012	0.054	0.007	0.017	0.003	0.036	0.023	0.007

Table 7 Euclidean distance from best and worst, P score and ranks

Candidate Site	Sİ +	Sİ-	P SCORE	RANK
1	0.03017	0.04761	0.61212	2
2	0.03895	0.03682	0.48591	4
3	0.03514	0.03556	0.50299	3
4	0.02228	0.04902	0.68749	1

study region. The region, which is closer to Konya city center compared to the other regions, differs from them due to its proximity to two different power transmission lines. Candidate Region 4 and its alternative, Candidate Region 1, determined according to the TOPSIS results, give an idea about the selection of a thermal power plant to be installed in the future.

Discussion

In the evaluation based on the application results within the scope of the study, spatial factors related to the thermal power plant installation area were primarily examined. Among the evaluated parameters, it was concluded that sinkhole areas (0.2579) were the most influential criterion. The most significant characteristic of the study area is the high presence of sinkholes and the high risk of new sinkhole formation. Due to this feature, the region encompasses adverse factors for both mining activities and facility installations. It is important to identify possible risks and identify sinkhole development and collapse mechanisms in terms of risk management. There are many studies in the literature that include sinkhole formation mechanisms and risk analyzes for different regions (Fidelibus et al., 2011; Shalev and Lyakhovsky, 2012; Dursun, 2022). Especially in recent years, studies on creating sinkhole risk maps using methods such as GIS-AHP have attracted attention. Taheri et al. (2015) produced a sinkhole sensitivity model in a GIS environment by applying the analytic hierarchy process (AHP) approach and considering eight factors, including distance to faults, drop in water level, and use of groundwater. The results of this study indicate that AHP can be a useful approach for sensitivity assessment if data on the main control factors have sufficient accuracy and spatial coverage. Differently, in this study, it has been shown that the combined of BMW and TOPSIS methods can be used in selecting the most suitable site for a critical facility such as a thermal power plant in a risky area containing sinkholes.

The second influential criterion was found to be the groundwater level. Sinkhole formation is directly or indirectly related to the hydrogeological structure. Groundwater plays an important role in sinkhole development, both as the agent responsible for dissolution and as the main factor determining ground stability. There are many articles examining the impact of the decrease in groundwater level on sinkhole formation (Taheri et al., 2015; Youssef et al., 2016; Orhan, 2021). Recent studies reveal that the majority of sinkholes are caused by changes in hydrogeological systems (Waltham et al., 2005; Acero et al., 2015; Parise et al., 2015; Gutiérrez, 2016; Linares et al., 2017). Therefore hydrogeological data obtained for the region with risk are essential for sinkhole risk assessment.In their study conducted in Iran, Siefi et al. (2017) used AHP and GIS together to select a location for a thermal power plant. As a different approach, they employed the WLC method from AHP techniques. With this method, the best site can be selected based on parameters such as cost, availability, and accessibility. A total of 8 criteria were evaluated under environmental and economic main categories. Through weighting with WLC, criteria with similar weights in terms of effectiveness were distance to settlements, fault lines and the groundwater level. As a result of the article, it was determined that the groundwater level is an important factor in the installation of thermal power plants.

The third most influential criterion is land use. The need for water is constantly increasing due to some factors such as rapid population growth, urbanization/industrialization, and changes in agricultural policies (Orhan, 2021; Yang et al., 2019). Therefore, land use can be seen as a directly effective factor in terms of groundwater use. In regions where surface water is not available, groundwater is used to irrigate agricultural lands. Lowering the groundwater levels affects the frequency of sinkhole formation. Fluctuations in groundwater levels, along with the overuse of groundwater aquifers, are often triggering factors in land subsidence and sinkhole development. Indirectly, it is understood that this criterion's high weightage is related to sinkholes. Therefore, it has been concluded that sinkholes and the top three criteria that affect sinkhole formation are significantly more important in thermal power plant site selection, accounting for a total of 52% weightage, compared to other factors.

It is understood that the fourth and fifth most important criteria are respectively the distance to fault lines and the Earthquake Epicenter and Magnitude. When choosing a power plant location, it is very important to consider seismic factors that significantly affect the operational safety risks of the power plant and its cost (Abdullah et al., 2023; Agyekum et al., 2021). In the literature, it is seen that the criterion of distance to fault lines is frequently considered in site selection studies for power plants. Baskurt and Aydın (2018) and Siefi et al. (2017) in choosing the location of nuclear power plant and thermal power plants respectively, used the distance to fault lines as a criterion in their study on the subject that earthquakes can cause emergencies and surface movements directly affect the facilities. Başkurt applied buffer analysis to seismic activity points. In this study, density analysis was also performed according to the intensity of seismic activities and buffer analysis was performed for the distance to fault lines. Abdullah et al. (2023), determined that among the criteria prioritized in the evaluation for the selection of a nuclear power plant using Fuzzy AHP, Geological, geotechnical, and seismological criteria are among the criteria with the highest priority. Although seismicity cannot be directly correlated with sinkhole formation, it can be considered that surface vibrations from earthquakes might accelerate sinkhole formation.

After determining the criteria in site selection studies in power plants, MCDM methods are applied to determine the weights of each criterion relative to other criteria. In the literature, there are many studies conducted with AHP, AHP-TOPSIS, Fuzzy AHP-TOPSIS and ANP methods (Colak et al., 2020; Sanchez-Lozano et al., 2013; Wang et al., 2018; Nasehi et al., 2016). Considering the studies carried out for the thermal power plant, Siefi et al., in 2017, weighted the criteria by applying the AHP method. As a result, they obtained the most suitable region in a very wide area. Choudhary & Shankar, 2012 made criterion weighting with fuzzy AHP and 5 alternative sites suggested were evaluated with TOPSIS. The fact that the alternative site is limited (compared to broad or previously determined alternatives) will not produce a sound solution for decision makers in power plant installation. Investor/s will prefer approaches that produce clearer results.

In the literature, it is seen that the Best–Worst method has not been used in criterion impact rating before in thermal power plant location selection studies. BWM-TOPSIS methods were used in the study in order to contribute to the literature and present a new methodology. In the study, the four most suitable alternative sites were determined by integrating them into GIS with BWM method weights. Within the scope of the study, this approach shows that the alternative sites appear in different areas according to the locational characteristics of the region, as a result of the weighting of BWM, and that it demonstrates a methodologically correct approach. The TOPSIS method was used to select the closest to the ideal solution of these fields.

Studies for the establishment of a thermal power plant in the region have been started. However, the project was canceled because the region was found to be high risk within the scope of Environmental Impact Assessment (EIA).However, lignite reserves in the region are of great importance for the country's economy. In the 11th Development Plans in Turkiye, "The use of our lignite reserves in the production of electrical energy in accordance with environmental standards will be increased." and "By ensuring that the lignite fields in public hands are brought into the economy through electricity generation, dependence on imported resources in electricity generation will be reduced and employment will be contributed." is presented as a target (11th Development Plan, 2019). In line with these objectives using the existing reserves of the region will be a good motivation for achieving the targets, both to meet the economic need for energy with local production and to reduce the 90% foreign-dependent energy need. The selection of locations for thermal power plants is a strategic decision that significantly influences the efficiency and production costs of these plants, concurrently impacting the sustainable development of countries (Choudhary & Shankar, 2012). The site selection process discussed in the study for a thermal power plant will establish an infrastructure for choosing different facility locations in regions bearing similar risks. It is believed that the methodology proposed in this study could provide insights into harnessing natural resources that remain untapped in the global regions due to similar risks, contributing to their integration into the economy.

Conclusion

In this study, 12 criteria were identified under geological, economic, and environmental main categories. Various analyses were applied to these criteria using a GIS software, ArcGIS. The analysis results were visualized through maps and scored accordingly. Criteria weights were determined by decision makers using BWM. According to the BWM results, sinkhole density criterion has been determined as the most weighted criterion. Following the sinkhole density criterion are groundwater level and land use criteria. A suitability map was created with the result weights. In the suitability map, it was seen that 10% of the 46 km^2 of the study area constituted suitable regions. The regions that appeared as suitable regions were selected as candidate regions on the map. In total, 4 candidate regions were identified. TOPSIS method was used to rank these 4 candidate regions in terms of suitability and to select the most suitable region. As a result of the application of the method, it was concluded that Candidate Zone 4, which has an area of 19 km^2, is the most suitable location for the thermal power plant. According to the results of the feasibility studies to be carried out on this area, it can be made ready for the installation of the power plant.

The most important originality of the study in this article among the studies on the installation of thermal power plants is that the study area is located in the region with the highest risk of sinkhole formation in Turkey. Of course, a thermal power plant to be established in such a region involves various dangers. However, the most important factor that encourages the study in the region is the richness of the coal reserve resource that can be used in the thermal power plant and the suitable of the coal. Due to the magnitude of the investment, risks posed by the region, and the time-consuming and costly nature of the project, it is essential that site selection for the facility in the field of energy production be carried out by experts using advanced methods. Therefore, a comprehensive analysis, incorporating methods that consider all factors directly or indirectly influencing facility location selection, is of paramount importance.

The result maps obtained in this study can serve as a base for many other studies. For example, with the support of the country in renewable energy sources, solar power plants have been established in this region. The result map, considering risk formation and spatial factors for these plants, can be used as a base map. Furthermore, the spatial evaluation of the facilities required for investment activities in the region can be assessed using the map generated from this study. Choosing a suitable location for thermal power plants requires taking into account and evaluating many complex parameters together. The combined use of BMW and TOPSIS methods provides a significant convenience in ensuring optimum location selection by correctly evaluating the relationship between these parameters. The results obtained from this study will provide insight into the application of similar methods in different regions and facilities.

Author contributions Ali Dogan Gumussoy: Data curation,Conceptualization, Methodology; Vildan Onen: Writing-Reviewing and Editing, Validation; Sukran Yalpir: Writing-Original draft preparation, Visualization, Investigation.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent to publish Not applicable.

Competing interests The authors declare no conflicts of interest.

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